Beyond the Triangle: Applications of Trigonometry

Translating the Real World into Geometry

Prepared by Harshit Jain | Delhi Public School (DPS) 67A Gurugram | Class X Mathematics

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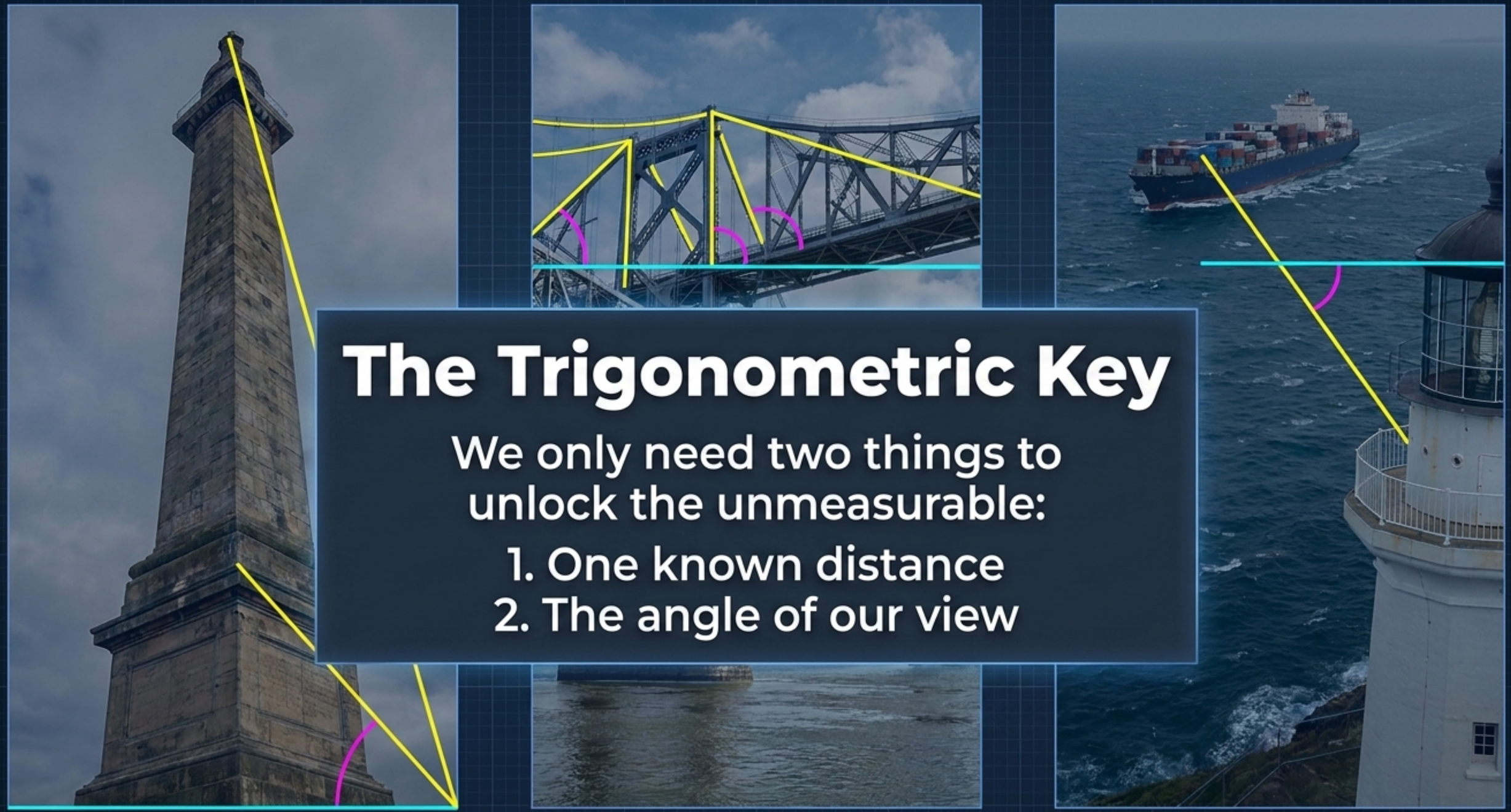

The Trigonometric Key

We only need two things to unlock the unmeasurable:

1. One known distance
2. The angle of our view

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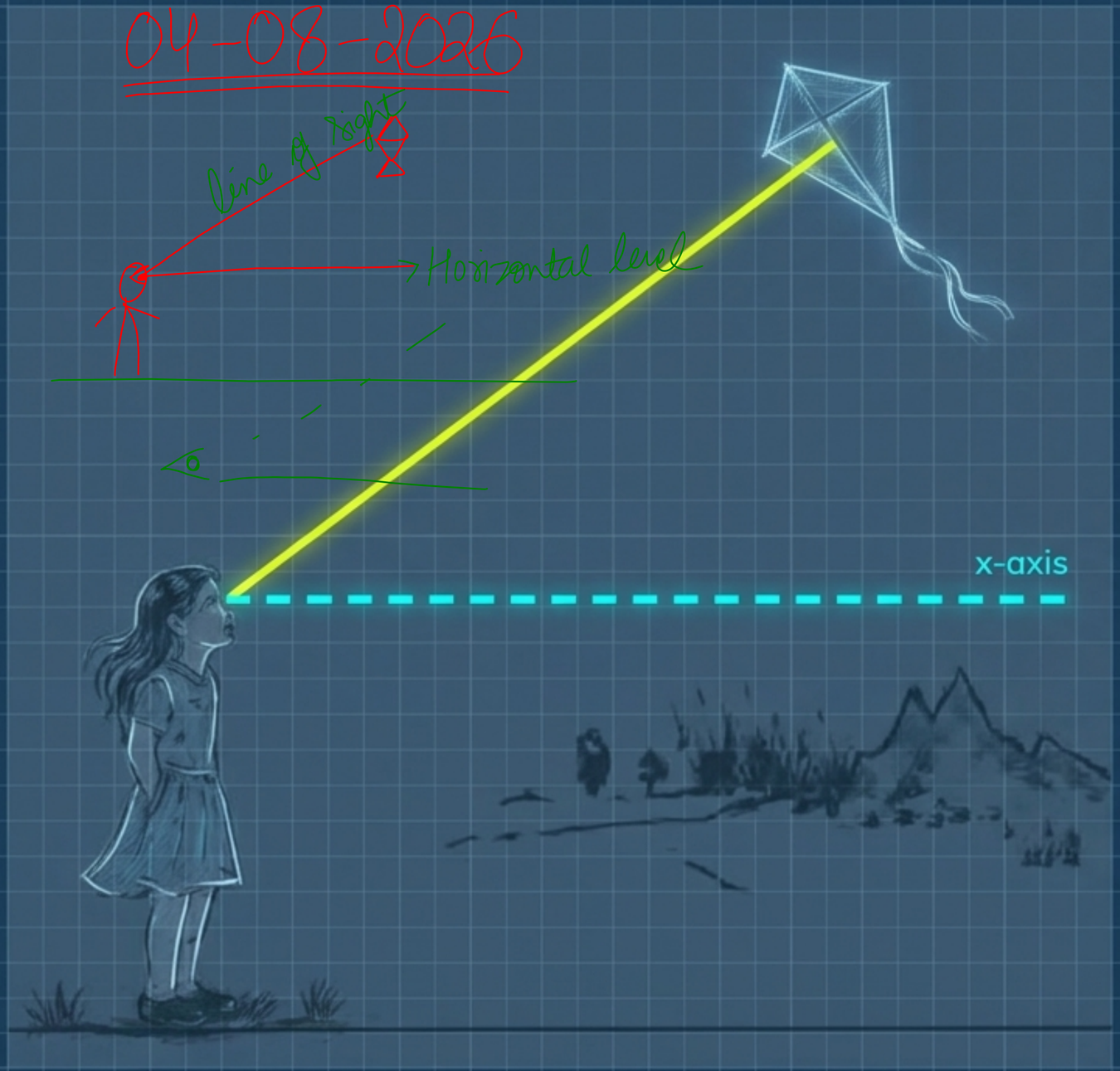

Mapping the Observer

Horizontal Level: The anchor line drawn straight ahead from the observer's eye, parallel to the ground.

Line of Sight: The direct line drawn from the eye of the observer to the point being viewed.

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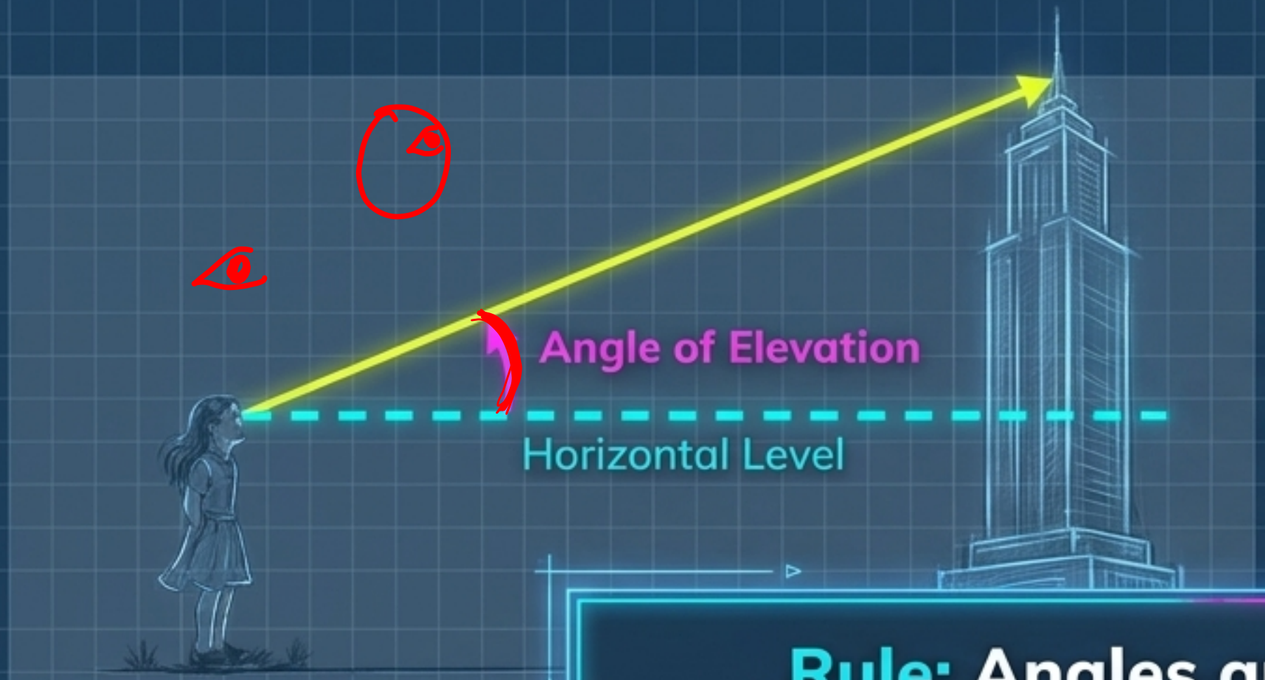
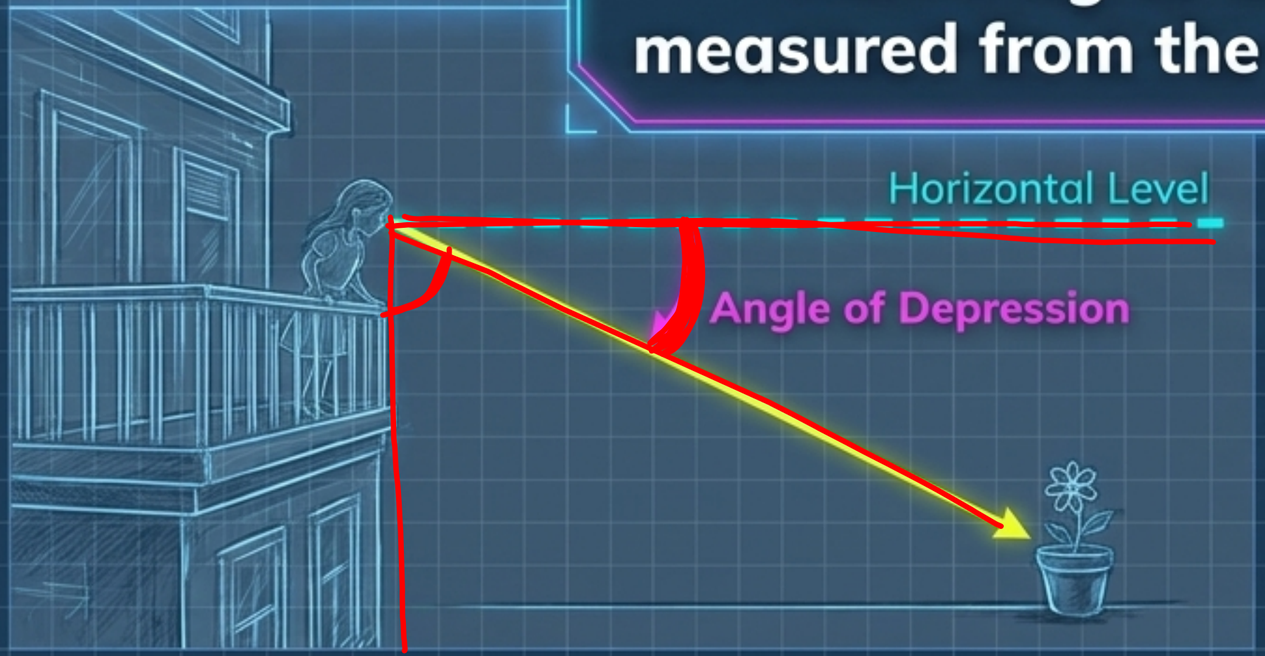


Angle of Elevation (Looking Up). Formed when the viewed object is above the horizontal level.

Rule: Angles are ALWAYS measured from the Horizontal Level.



Angle of Depression (Looking Down). Formed when the viewed object is below the horizontal level.

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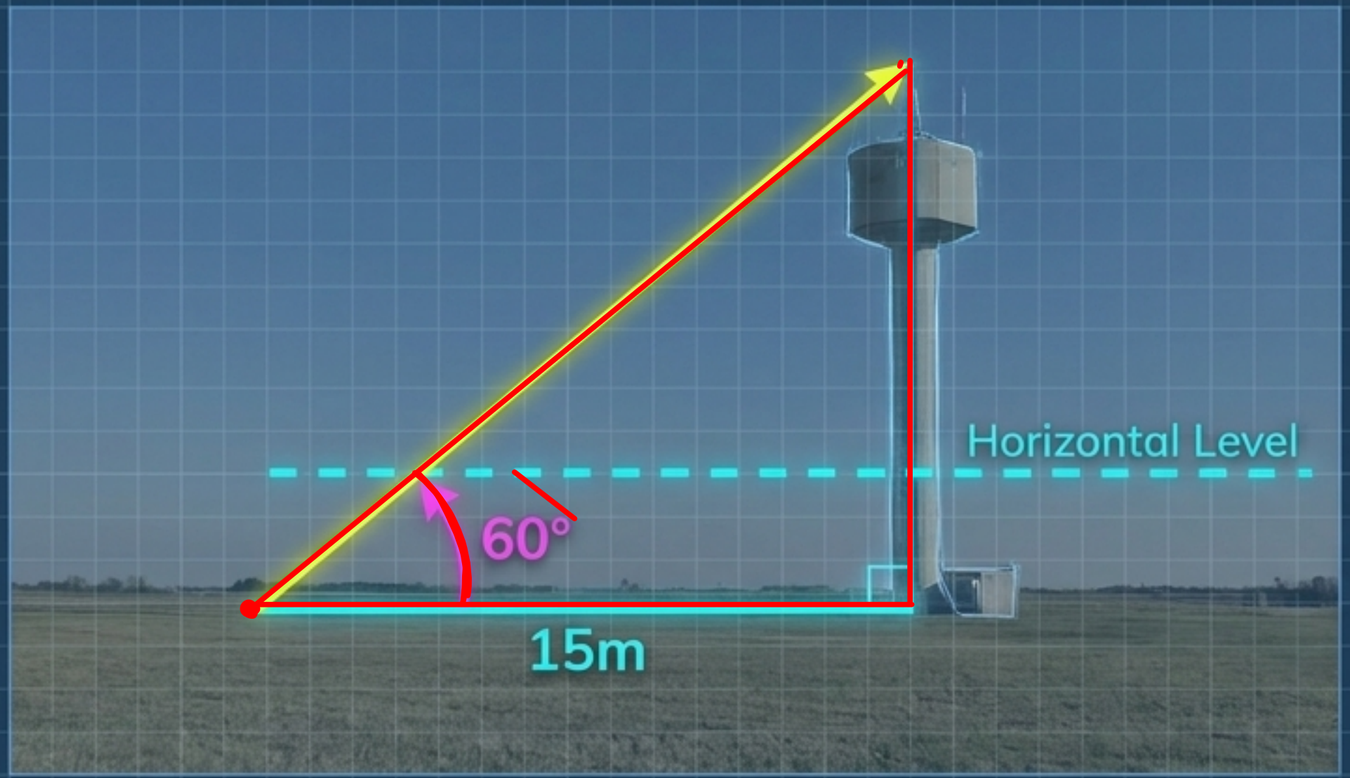
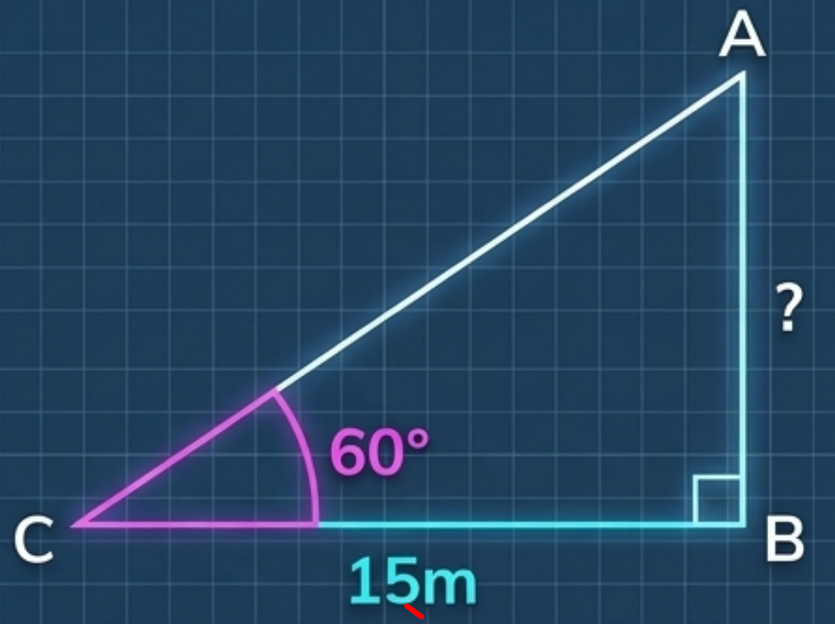
Example 1: The Simple Tower





Problem Details

Distance from foot = 15m.
Angle of elevation = 60°.
Find tower height (AB).

Math Breakdown

$$\tan(60°) = \frac{\text{Opposite}}{\text{Adjacent}} = \frac{AB}{BC}$$

$$\sqrt{3} = \frac{AB}{15}$$

$$AB = 15\sqrt{3} \text{ m.}$$

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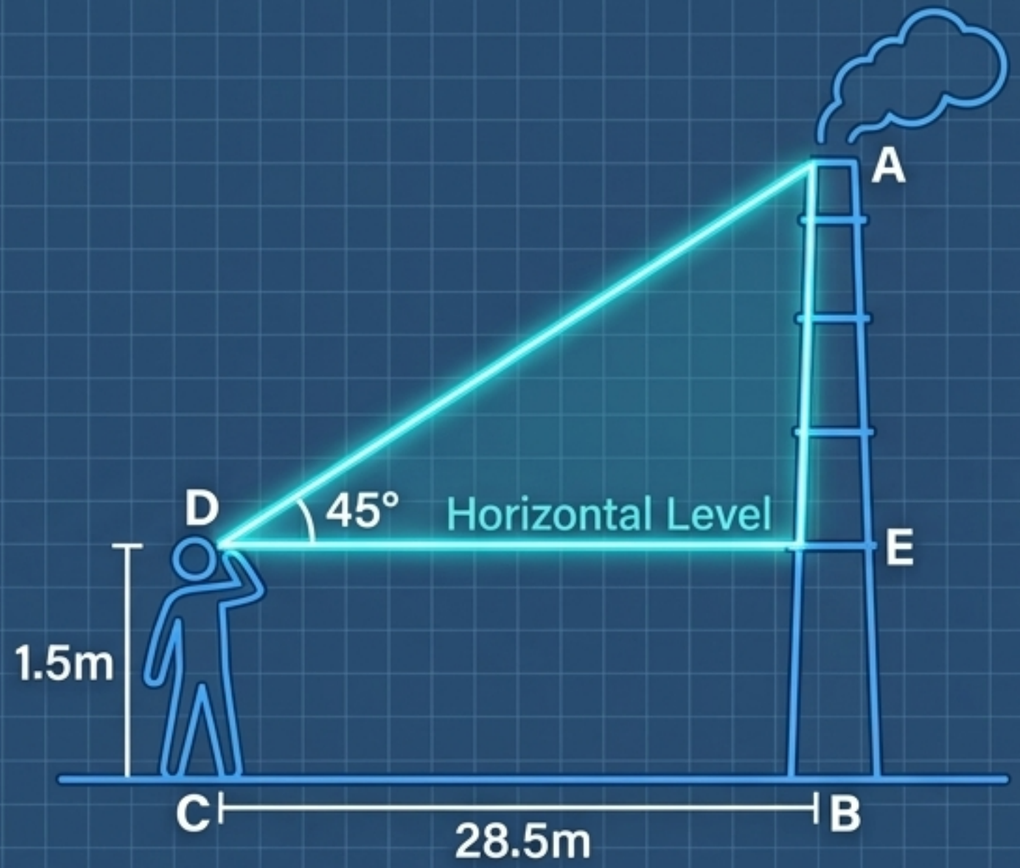
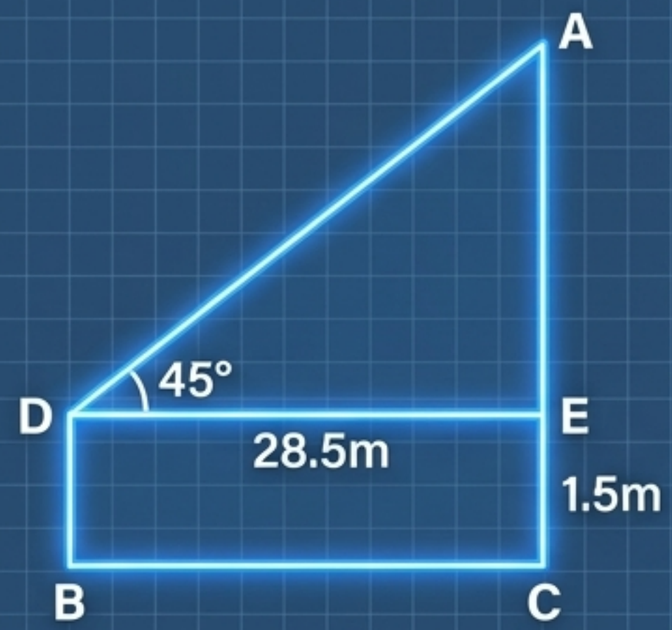
Example 3: Accounting for the Observer



Problem Details

Observer (1.5m tall) is 28.5m away from chimney.
Angle of elevation = 45°.
Find chimney height.



Math Breakdown

tan(45°) = AE / 28.5
1 = AE / 28.5 → AE = 28.5 m.

Don’t forget the rectangle!

Total Height (AB) = AE (Triangle) + BE (Observer)
= 28.5 + 1.5 = 30 m.

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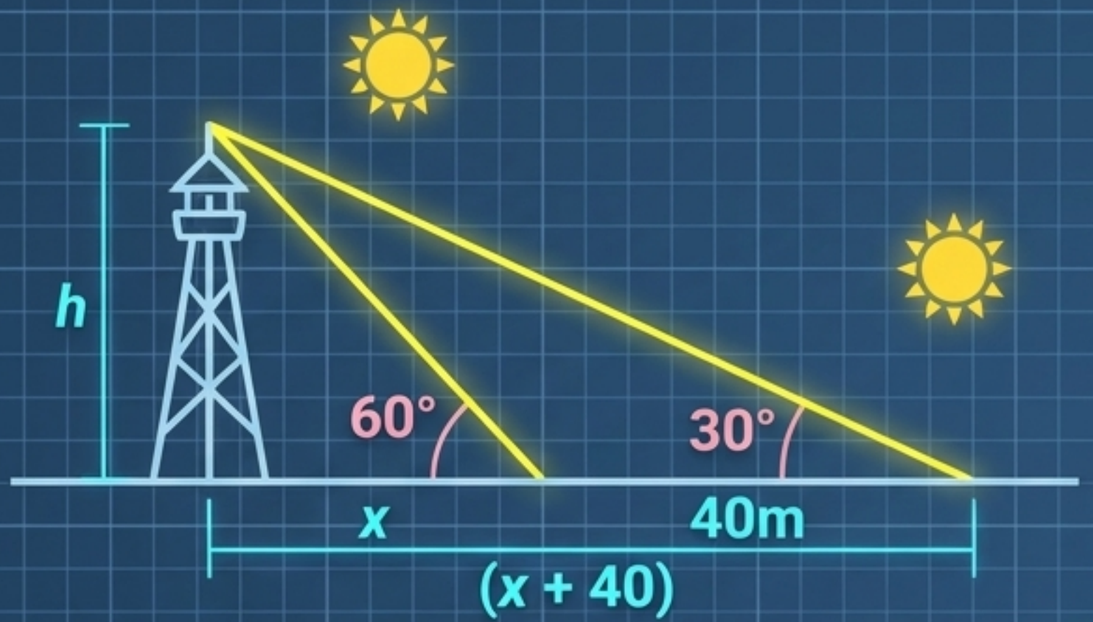
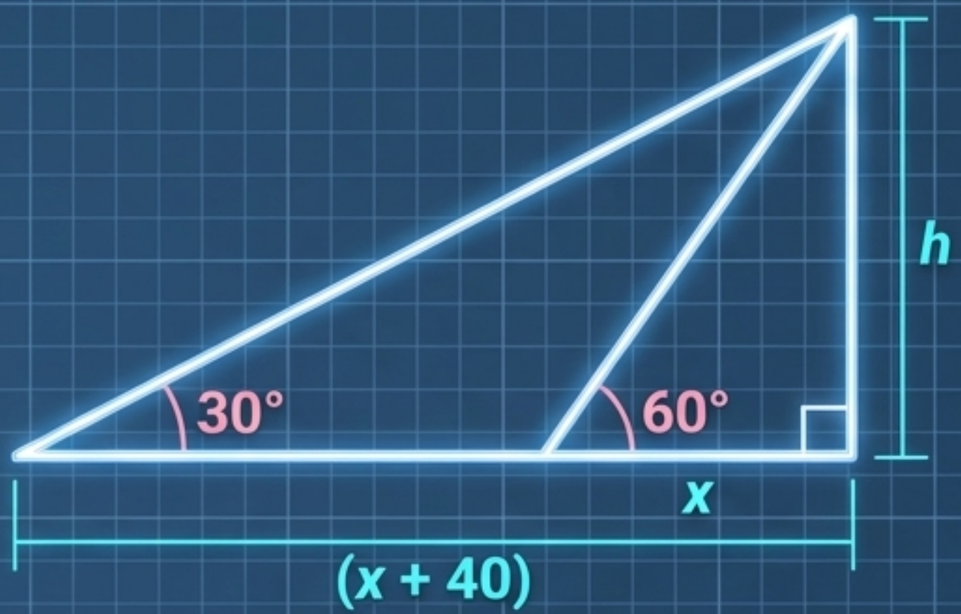
Example 5: The Shifting Shadow





Problem details: Shadow is 40m longer at 30° altitude than at 60°.

Dual-Column Math Layout

Column 1	Inner Triangle - 60°	Column 2	Outer Triangle - 30°
$\tan(60°) = \frac{h}{x} \rightarrow \sqrt{3} = \frac{h}{x} \rightarrow h = x\sqrt{3}$		$\tan(30°) = \frac{h}{x+40)} \rightarrow \frac{1}{\sqrt{3}} = \frac{h}{x+40}$	
Synthesis	Substitute h: $\sqrt{3}(x\sqrt{3}) = x + 40 \rightarrow 3x = x + 40 \rightarrow x = 20.$		

Result: Height (h) = $20\sqrt{3}$ m.

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EXERCISE 9.1

1. A circus artist is climbing a 20 m long rope, which is tightly stretched and tied from the top of a vertical pole to the ground. Find the height of the pole, if the angle made by the rope with the ground level is 30° (see Fig. 9.11).

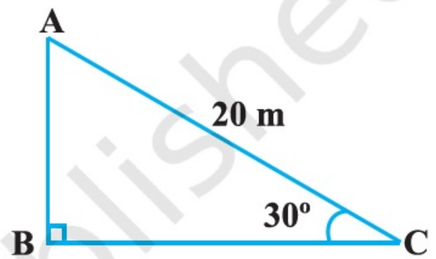
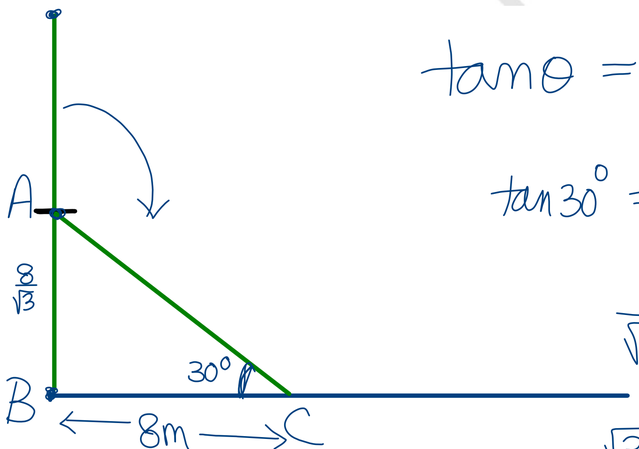


Fig. 9.11

2. A tree breaks due to storm and the broken part bends so that the top of the tree touches the ground making an angle 30° with it. The distance between the foot of the tree to the point where the top touches the ground is 8 m. Find the height of the tree.



$$\tan\theta = \frac{AB}{BC}$$

$$\tan 30^\circ = \frac{AB}{8}$$

$$\frac{1}{\sqrt{3}} = \frac{AB}{8}$$

$$\sqrt{3}\cdot AB = 8$$

$$AB = \frac{8}{\sqrt{3}} = \frac{8\sqrt{3}}{3}\ m$$

$$\cos\theta = \frac{BC}{AC}$$

$$\cos 30^\circ = \frac{8}{AC}$$

$$\frac{\sqrt{3}}{2} = \frac{8}{AC}$$

$$16 = \sqrt{3}\cdot AC$$

$$AC = \frac{16}{\sqrt{3}} = \frac{16\sqrt{3}}{3}\ m$$

Height of tree $= AB + AC$

$$= \frac{8\sqrt{3}}{3} + \frac{16\sqrt{3}}{3} = \frac{24\sqrt{3}}{3} = 8\sqrt{3}\ m$$

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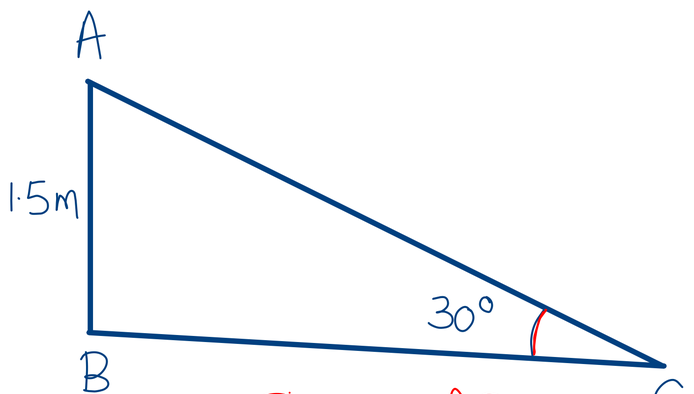
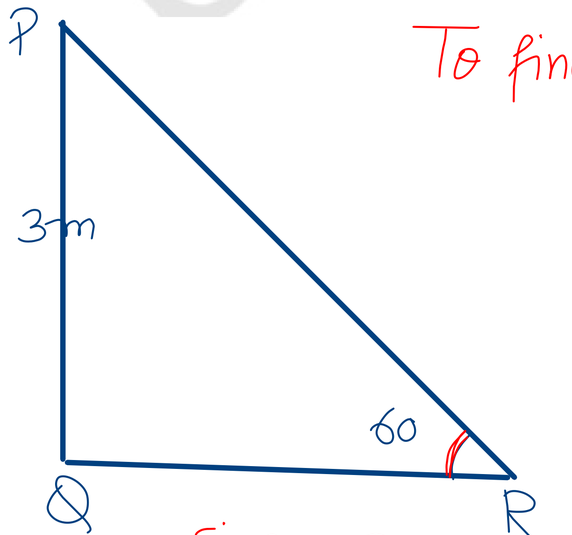
3. A contractor plans to install two slides for the children to play in a park. For the children below the age of 5 years, she prefers to have a slide whose top is at a height of 1.5 m, and is inclined at an angle of 30° to the ground, whereas for elder children, she wants to have a steep slide at a height of 3m, and inclined at an angle of 60° to the ground. What should be the length of the slide in each case?





To find : AC and PR

$\sin\theta = \frac{AB}{AC}$

$\sin 30 = \frac{1.5}{AC}$

$\frac{1}{2} = \frac{1.5}{AC}$

$\Rightarrow AC = 3\text{m}$

$\sin\theta = \frac{PQ}{PR}$

$\sin 60 = \frac{3}{PR}$

$\frac{\sqrt{3}}{2} = \frac{3}{PR}$

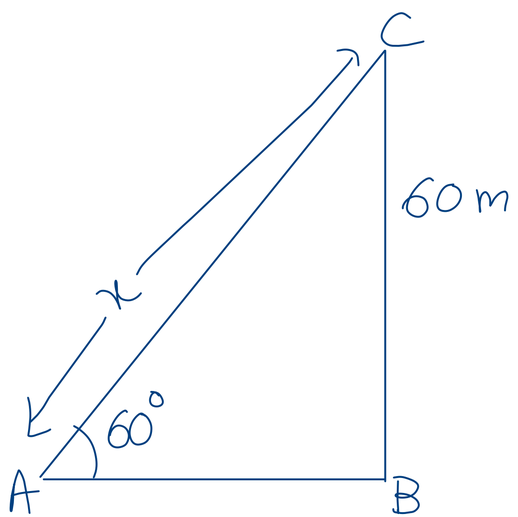
$\sqrt{3}PR = 6$

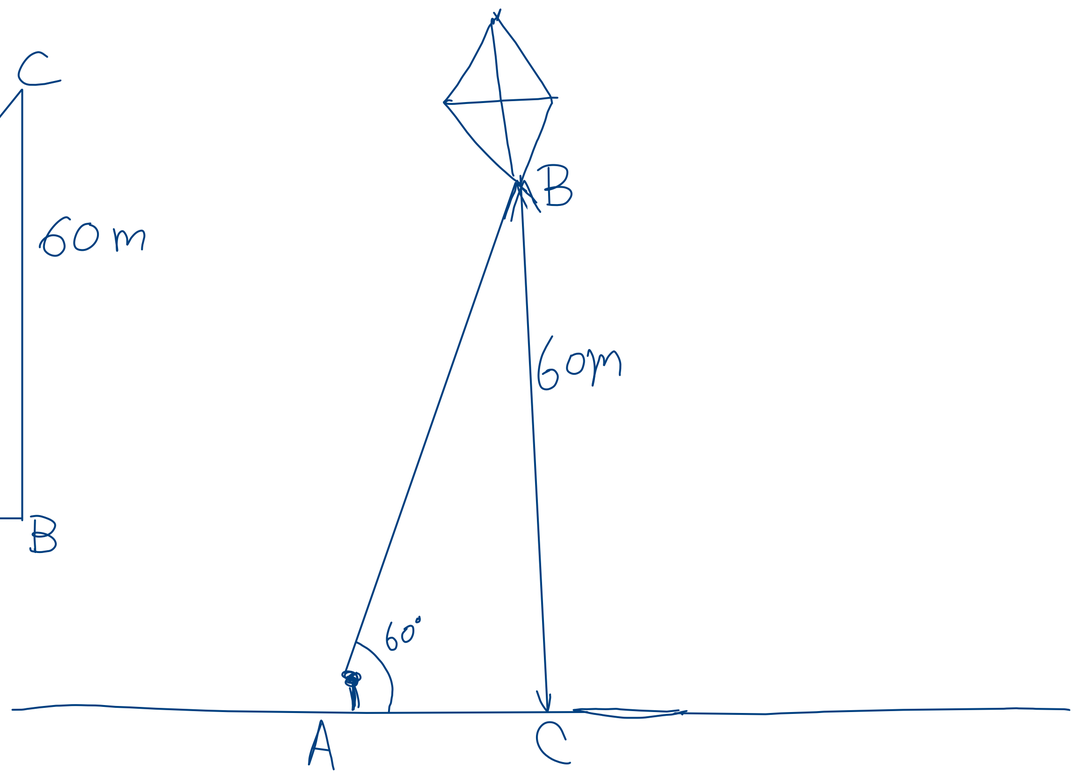
$PR = \frac{6}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$

$= \frac{6\sqrt{3}}{3} = \boxed{2\sqrt{3}\text{ m}}$

4. The angle of elevation of the top of a tower from a point on the ground, which is 30 m away from the foot of the tower, is 30°. Find the height of the tower.

5. A kite is flying at a height of 60 m above the ground. The string attached to the kite is temporarily tied to a point on the ground. The inclination of the string with the ground is 60°. Find the length of the string, assuming that there is no slack in the string.





$\sin 60° = \frac{60}{x}$

$\frac{\sqrt{3}}{2} = \frac{60}{x}$

$x = 120 \times \frac{\sqrt{3}}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{120\sqrt{3}}{3} = 40\sqrt{3}$ m

6. A 1.5 m tall boy is standing at some distance from a 30 m tall building. The angle of elevation from his eyes to the top of the building increases from 30° to 60° as he walks towards the building. Find the distance he walked towards the building.

7. From a point on the ground, the angles of elevation of the bottom and the top of a transmission tower fixed at the top of a 20 m high building are 45° and 60° respectively. Find the height of the tower.

In $\triangle ABC$

$$\tan\theta = \frac{BC}{AB}$$

$$\tan 45^\circ = \frac{20}{AB}$$

$$1 = \frac{20}{AB}$$

$$AB = 20 \text{ m}$$

In $\triangle ABD$

$$\tan\theta = \frac{DB}{BA}$$

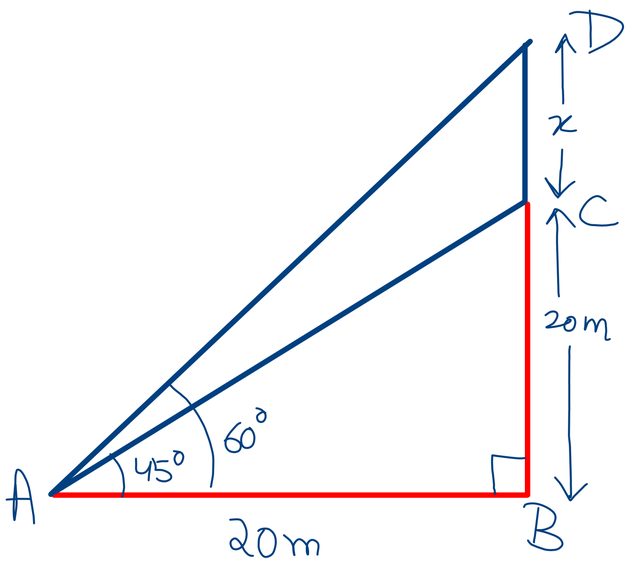
$$\tan 60^\circ = \frac{20 + x}{20}$$

$$\sqrt{3} = \frac{20 + x}{20}$$

$$20 + x = 20\sqrt{3}$$

$$x = 20\sqrt{3} - 20$$

$$x = 20(\sqrt{3} - 1)$$



8. A statue, 1.6 m tall, stands on the top of a pedestal. From a point on the ground, the angle of elevation of the top of the statue is 60° and from the same point the angle of elevation of the top of the pedestal is 45°. Find the height of the pedestal.

9. The angle of elevation of the top of a building from the foot of the tower is 30° and the angle of elevation of the top of the tower from the foot of the building is 60°. If the tower is 50 m high, find the height of the building.

In $\triangle BDC$

$\tan\theta = \frac{DC}{BC}$

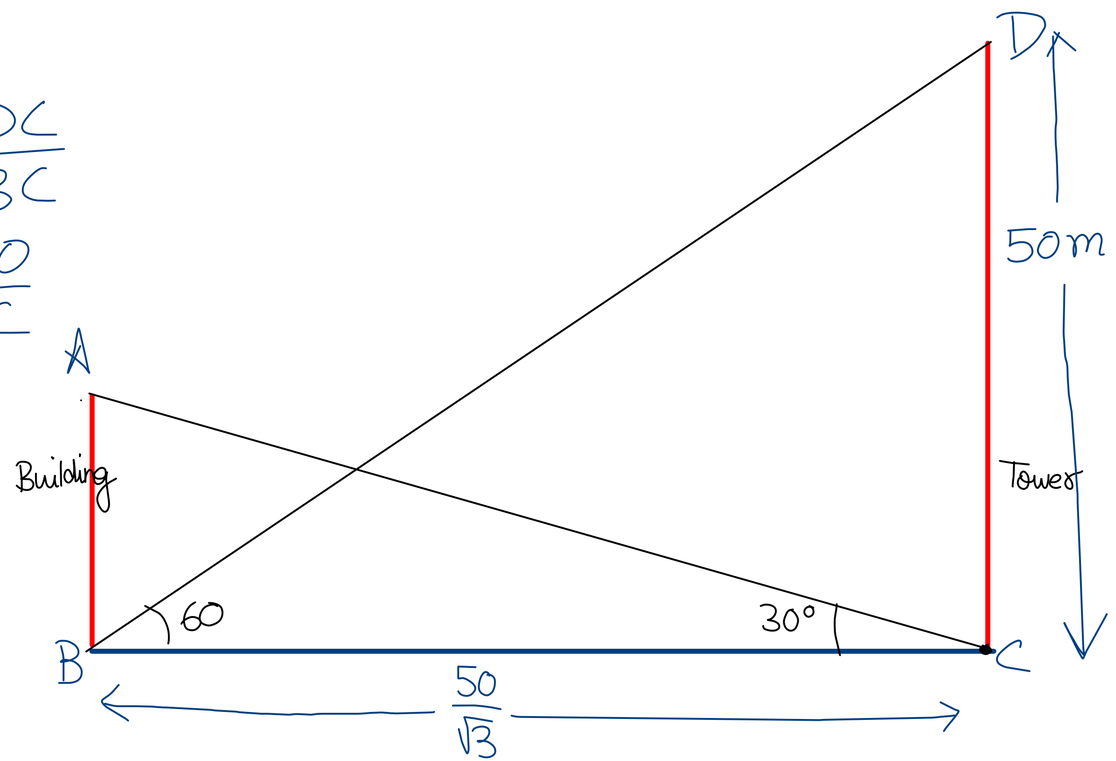
$\tan 60 = \frac{50}{BC}$

$\sqrt{3} = \frac{50}{BC}$

$\sqrt{3}BC = 50$

$BC = \frac{50}{\sqrt{3}}$

$=$



In $\triangle ABC$

$\tan 30^\circ = \frac{AB}{BC}$

$\frac{1}{\sqrt{3}} = \frac{AB}{\left(\frac{50}{\sqrt{3}}\right)}$

$\frac{1}{\sqrt{3}} = \frac{AB\sqrt{3}}{50}$

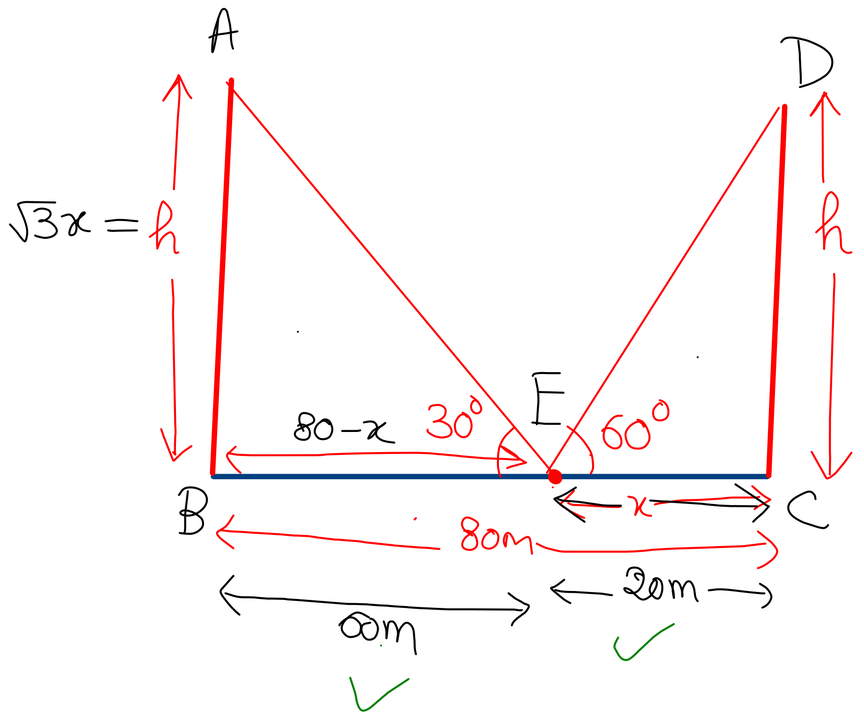
$3AB = 50$

$AB = \frac{50}{3} = 16.\overline{6}$

≈ 16.67 m

$= 16\frac{2}{3}$ m

10. Two poles of equal heights are standing opposite each other on either side of the road, which is 80 m wide. From a point between them on the road, the angles of elevation of the top of the poles are 60° and 30°, respectively. Find the height of the poles and the distances of the point from the poles.



In ΔDEC

$\tan 60^\circ = \frac{h}{x}$

$\sqrt{3} = \frac{h}{x}$

$\sqrt{3}x = h$

$h = \sqrt{3} \times 20$

$= 20\sqrt{3}$ m

In ΔABE

$\tan 30^\circ = \frac{\sqrt{3}x}{80-x}$

$\frac{1}{\sqrt{3}} = \frac{\sqrt{3}x}{80-x}$

$\Rightarrow 3x = 80 - x$

$\Rightarrow 4x = 80$

$\Rightarrow x = 20$

11. A TV tower stands vertically on a bank of a canal. From a point on the other bank directly opposite the tower, the angle of elevation of the top of the tower is 60°. From another point 20 m away from this point on the line joing this point to the foot of the tower, the angle of elevation of the top of the tower is 30° (see Fig. 9.12). Find the height of the tower and the width of the canal.

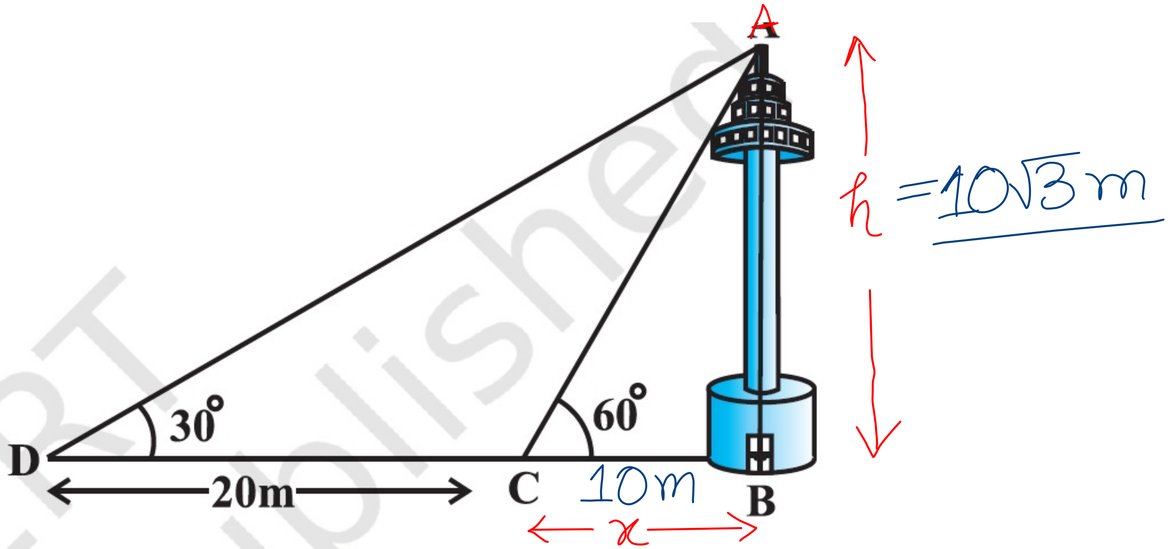


Fig. 9.12

In ΔADB

$\tan 30^\circ = \frac{h}{20+x}$

$\frac{1}{\sqrt{3}} = \frac{h}{20+x}$

In ΔABC

$\tan 60^\circ = \frac{h}{x}$

$\sqrt{3} = \frac{h}{x}$

$h = \sqrt{3}x$

$h = \sqrt{3} \times 10$

$= 10\sqrt{3}$

$\frac{1}{\sqrt{3}} = \frac{\sqrt{3}x}{20+x}$

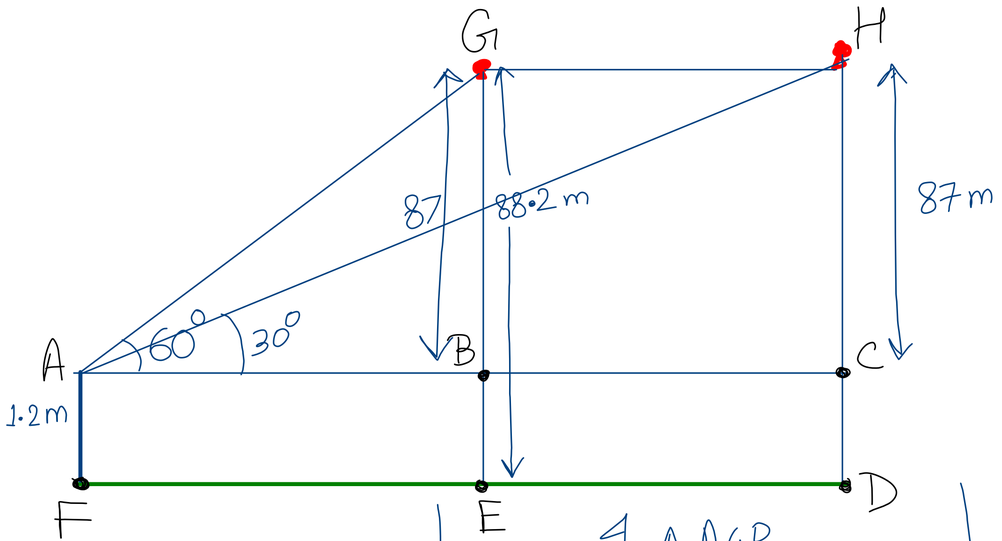
$3x = 20 + x$

$2x = 20$

$x = 10$

12. From the top of a 7 m high building, the angle of elevation of the top of a cable tower is 60° and the angle of depression of its foot is 45°. Determine the height of the tower.

13. As observed from the top of a 75 m high lighthouse from the sea-level, the angles of depression of two ships are 30° and 45°. If one ship is exactly behind the other on the same side of the lighthouse, find the distance between the two ships.



$GB = GE - BE$

$= 88.2 - 1.2$

$= 87$

In $\triangle AGB$

$\tan 60^\circ = \frac{GB}{AB}$

$\Rightarrow \sqrt{3} = \frac{87}{AB}$

$\Rightarrow AB = \frac{87}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{87\sqrt{3}}{3}$

$\boxed{AB = 29\sqrt{3}}$

14. A 1.2 m tall girl spots a balloon moving with the wind in a horizontal line at a height of 88.2 m from the ground. The angle of elevation of the balloon from the eyes of the girl at any instant is 60°. After some time, the angle of elevation reduces to 30° (see Fig. 9.13). Find the distance travelled by the balloon during the interval.

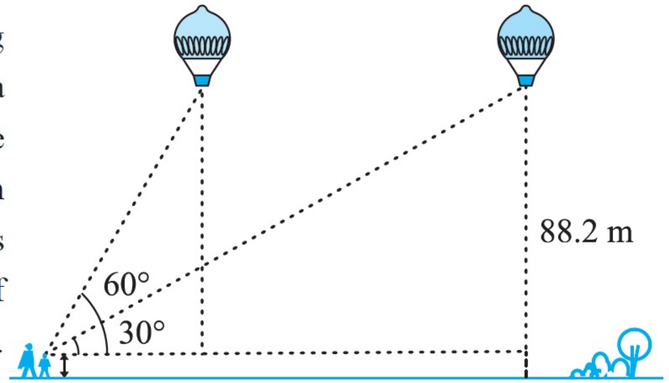


Fig. 9.13

In $\triangle AHC$

$\tan 30^\circ = \frac{HC}{AC} \Rightarrow \frac{1}{\sqrt{3}} = \frac{87}{AC}$

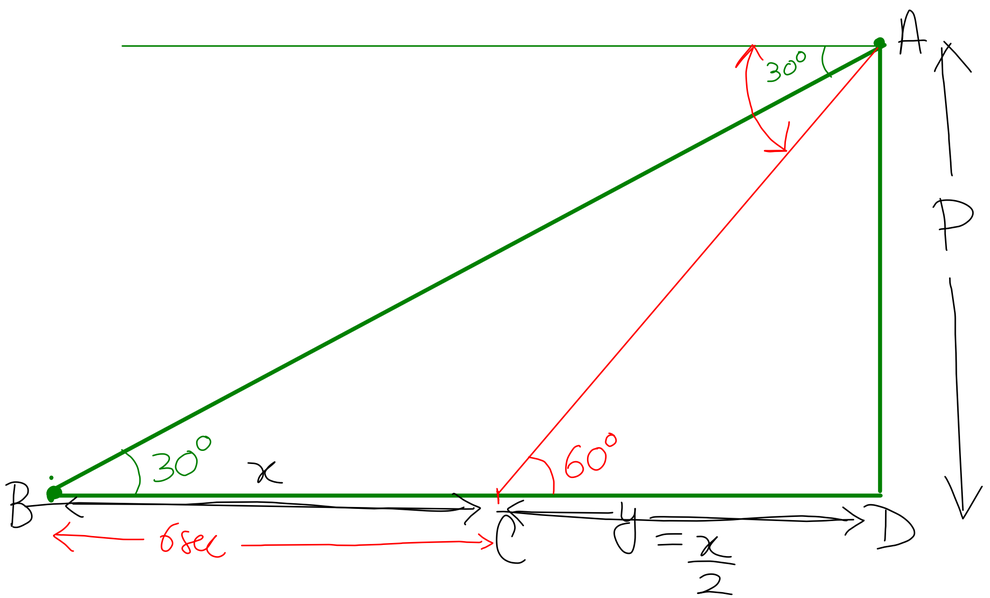
$\Rightarrow \boxed{AC = 87\sqrt{3}}$

$AC - AB = BC$

$87\sqrt{3} - 29\sqrt{3} = BC$

$\boxed{BC = 58\sqrt{3}\ m}$

15. A straight highway leads to the foot of a tower. A man standing at the top of the tower observes a car at an angle of depression of 30°, which is approaching the foot of the tower with a uniform speed. Six seconds later, the angle of depression of the car is found to be 60°. Find the time taken by the car to reach the foot of the tower from this point.



In $\triangle ABD$

$\tan 30 = \frac{AD}{BD}$

$\frac{1}{\sqrt{3}} = \frac{P}{BD}$

$BD = P\sqrt{3}$

$\Rightarrow BD = \sqrt{3}y \cdot \sqrt{3}$

$BD = 3y$

$x + y = 3y$

$x = 2y$

$y = \frac{x}{2}$

In $\triangle ACD$

$\tan 60^\circ = \frac{P}{y}$

$\sqrt{3} = \frac{P}{y}$

$P = \sqrt{3}y$

In $\triangle ACD$

$\tan 60^\circ = \frac{AD}{CD} = \frac{P}{y}$

$\sqrt{3} = \frac{P}{y}$

$\boxed{P = \sqrt{3}y}$

$y = \frac{P}{\sqrt{3}} = \frac{P\sqrt{3}}{3}$

In $\triangle ABD$

$\tan 30^\circ = \frac{AD}{BD}$

$\frac{1}{\sqrt{3}} = \frac{P}{x+y}$

$x + y = P\sqrt{3}$

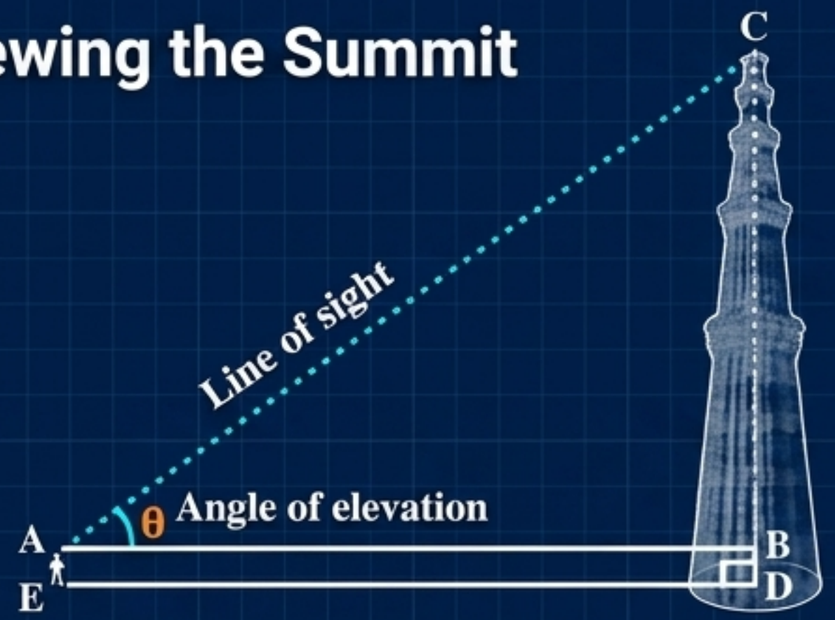
$x + \frac{P}{\sqrt{3}} = P\sqrt{3}$

$x + \frac{P\sqrt{3}}{3} = P\sqrt{3}$

$x = \frac{2P\sqrt{3}}{3}$

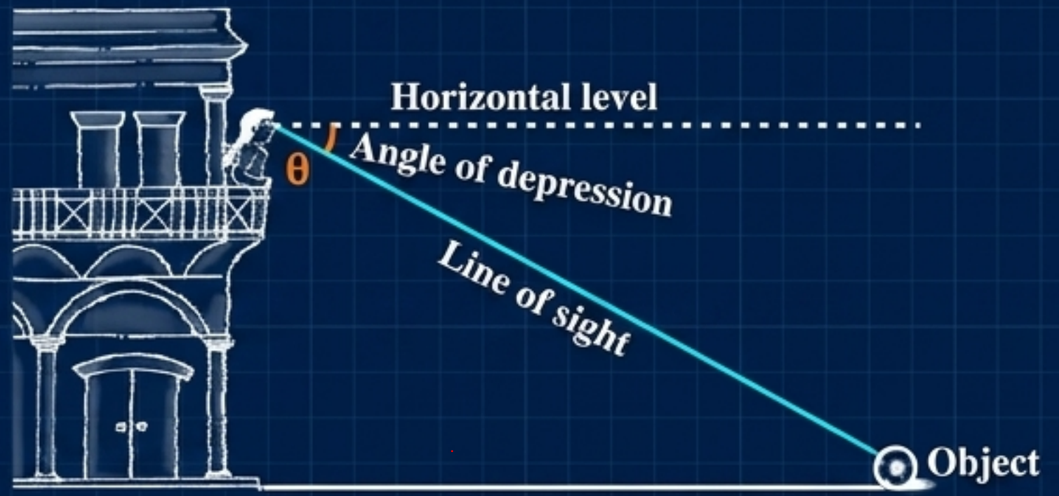
The Field Surveyor's Toolkit

Viewing the Summit



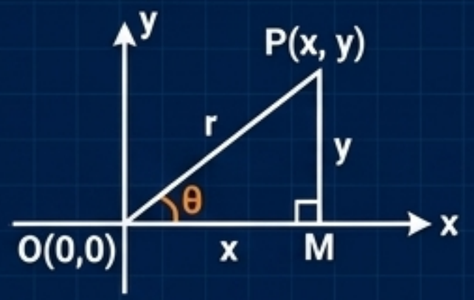
Line of sight **ABOVE** horizontal level. Head raised.

Viewing the Valley



Line of sight **BELOW** horizontal level. Head lowered.

The Golden Ratios



$$\sin\theta = \frac{\text{Opposite}}{\text{Hypotenuse}} = \frac{y}{r}$$

$$\cos\theta = \frac{\text{Adjacent}}{\text{Hypotenuse}} = \frac{x}{r}$$

$$\tan\theta = \frac{\text{Opposite}}{\text{Adjacent}} = \frac{y}{x}$$

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Phase I: Base Camp

Foundational Right Triangles

Establishing baseline measurements and single-angle calculations.

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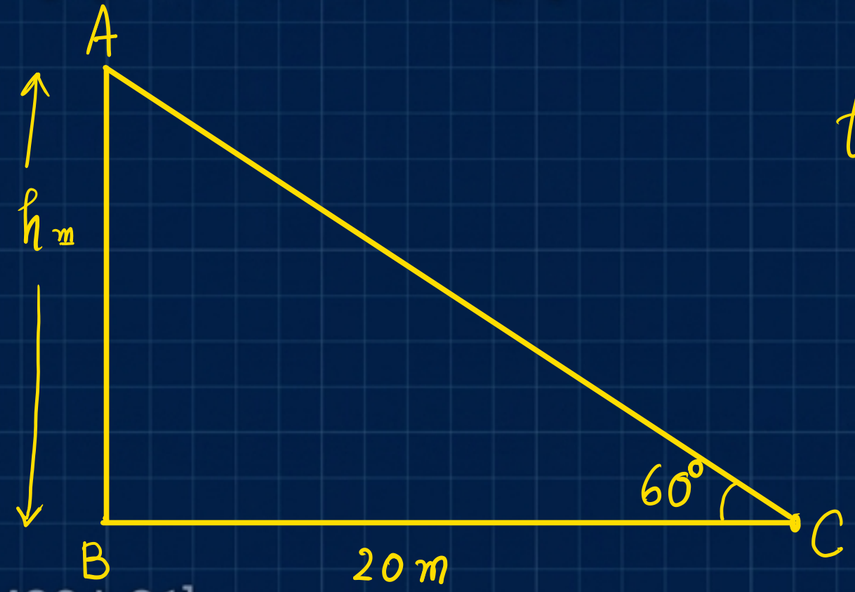
Field Target 01: The Tower

The angle of elevation of the top of a tower from a point on the ground, which is 20m away from the foot of the tower is 60°. Find the height of the tower.

(a) $10\sqrt{3}$ m (b) $30\sqrt{3}$ m (c) $20\sqrt{3}$ m (d) none of these



$\tan 60^\circ = \frac{AB}{BC}$

$\sqrt{3} = \frac{h}{20}$

$h = 20\sqrt{3}$ m

[MCQ I, Q1]

Correct Answer: (c) $20\sqrt{3}$ m

Key Step: $\tan(60^\circ) = \frac{h}{20} \rightarrow \sqrt{3} = \frac{h}{20}$

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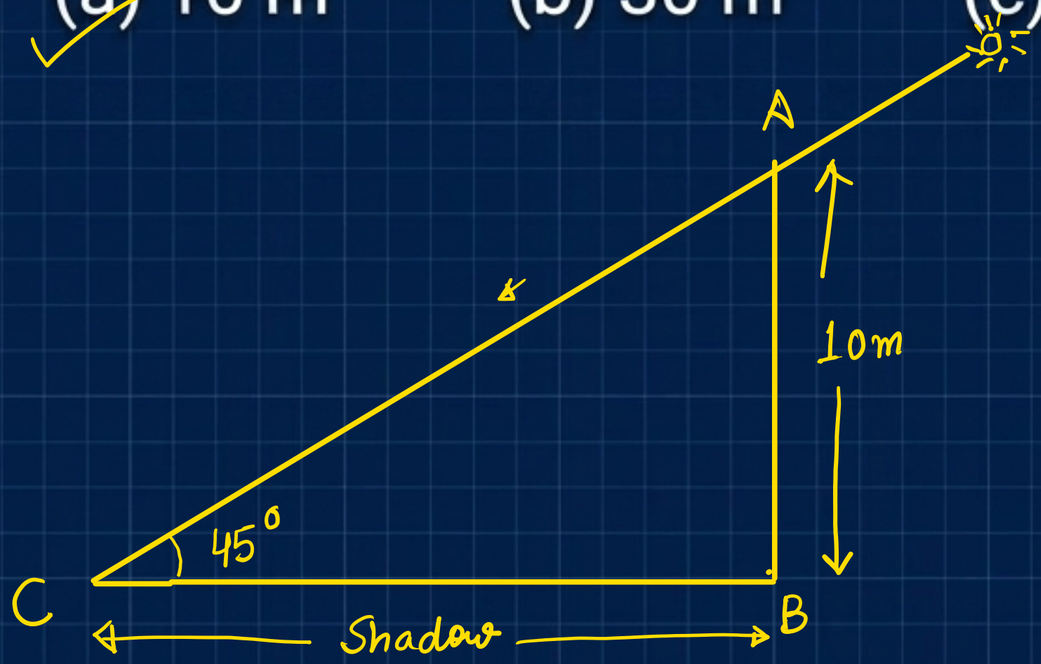
Field Target 02: The Shadow

The height of a tower is 10m. What is the length of its shadow when the Sun's altitude is 45°?

(a) 10 m (b) 30 m (c) 20 m (d) none of these



$\tan 45^\circ = \frac{AB}{BC}$

$1 = \frac{10}{BC}$

$BC = 10\ m$

[MCQ I, Q2]

Correct Answer: (a) 10 m

Key Step: $\tan(45^\circ) = \frac{10}{\text{shadow}} \rightarrow 1 = \frac{10}{\text{shadow}}$

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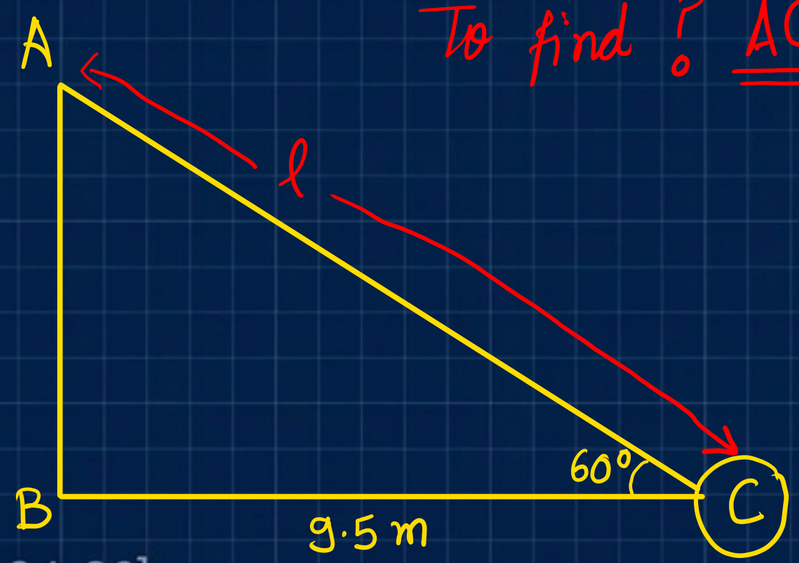
Field Target 03: The Ladder

The angle of elevation of a ladder leaning against a wall is 60° and the foot of the ladder is 9.5 m away from the wall. Find the length of the ladder.

(a) 10 m (b) 19 m (c) 20 m (d) none of these



To find ? AC

$\cos 60^\circ = \frac{BC}{AC}$

$\Rightarrow \frac{1}{2} = \frac{9.5}{l}$

$l = 9.5 \times 2$

$l = 19\ m$

[MCQ I, Q3]

Correct Answer: (b) 19 m

Key Step: $\cos(60^\circ) = \frac{9.5}{\text{length}} \rightarrow \frac{1}{2} = \frac{9.5}{\text{length}}$

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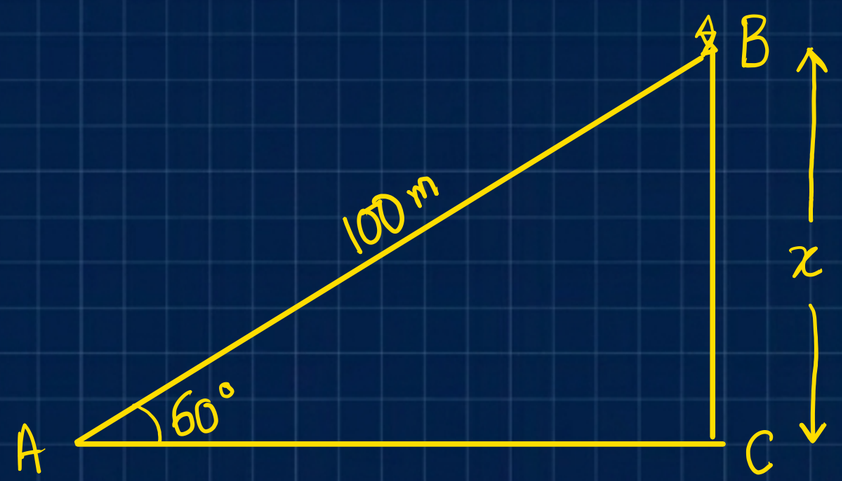
Field Target 04: The Kite

The string of a kite is 100m long and it makes an angle of 60° with the horizontal. Find the height of the kite, assuming that there is no slack in the string.

(a) $100\sqrt{3}$ m (b) $200\sqrt{3}$ m (c) $50\sqrt{3}$ m (d) $100/\sqrt{3}$ m



$\sin 60^\circ = \frac{x}{100}$

$\Rightarrow \frac{\sqrt{3}}{2} = \frac{x}{100}$

$\Rightarrow 2x = 100\sqrt{3}$

$x = \frac{100\sqrt{3}}{2}$

$x = 50\sqrt{3}$ m

[MCQ II, Q5]

Correct Answer: (c) $50\sqrt{3}$ m

Key Step: $\sin(60^\circ) = \frac{\text{height}}{100} \rightarrow \frac{\sqrt{3}}{2} = \frac{\text{height}}{100}$

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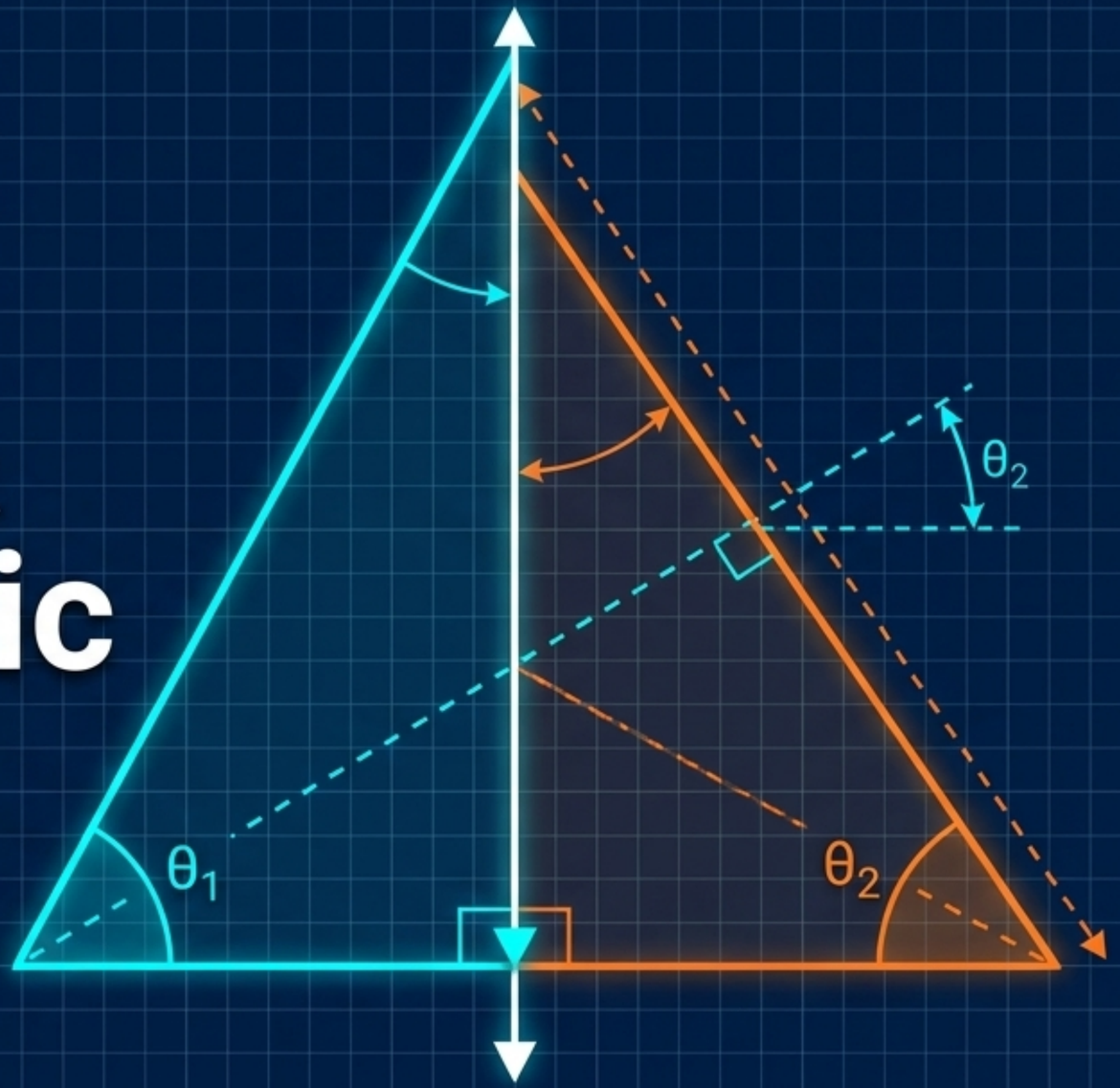
Phase II: The Ascent

Shifting Perspectives & Multi-Step Logic

Dual-angle problems requiring the synthesis of two overlapping overlapping right triangles.



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Ascent Protocol 01: Shifting Perspectives

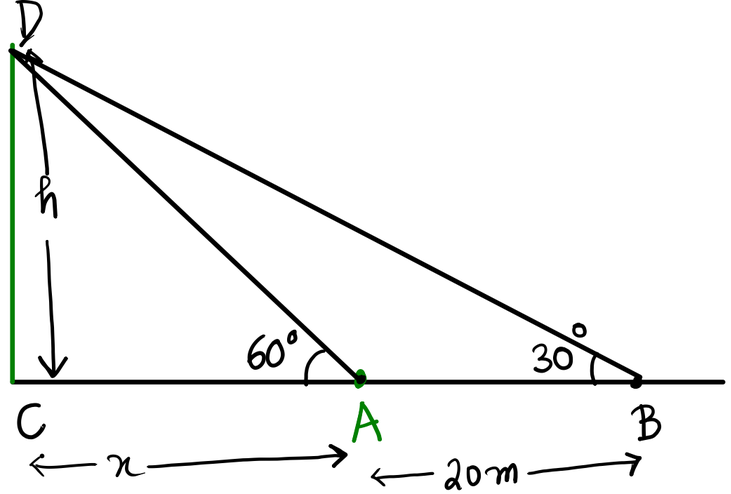
The angle of elevation of the top of a tree from a point A on the ground is 60°. On walking 20m away from its base, to a point B, the angle of elevation changes to 30°. Find the height of the tree.

(a) $10\sqrt{3}$ m (b) $30\sqrt{3}$ m
(c) $20\sqrt{3}$ m (d) none of these

[MCQ III, Q2]

Correct Answer: (a) $10\sqrt{3}$ m

Key Step: $h/x = \sqrt{3}$ AND $h/(x+20) = 1\sqrt{3} \rightarrow$ $\rightarrow 3x = x + 20 \rightarrow x = 10$. $h = 10\sqrt{3}$.



In ΔACD

$\tan 60° = \frac{h}{x}$

$\Rightarrow \sqrt{3} = \frac{h}{x}$

$h = \sqrt{3}x$

$h = \sqrt{3} \times 10$

Answer $h = 10\sqrt{3}$ m

In ΔBCD

$\tan 30° = \frac{h}{x+20}$

$\Rightarrow \frac{1}{\sqrt{3}} = \frac{h}{x+20}$

$\Rightarrow x + 20 = h\sqrt{3}$

$\Rightarrow x + 20 = \sqrt{3}x \times \sqrt{3}$

$x + 20 = 3x$

$2x = 20$

$x = 10$

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Ascent Protocol 02: Adjusting for the Observer

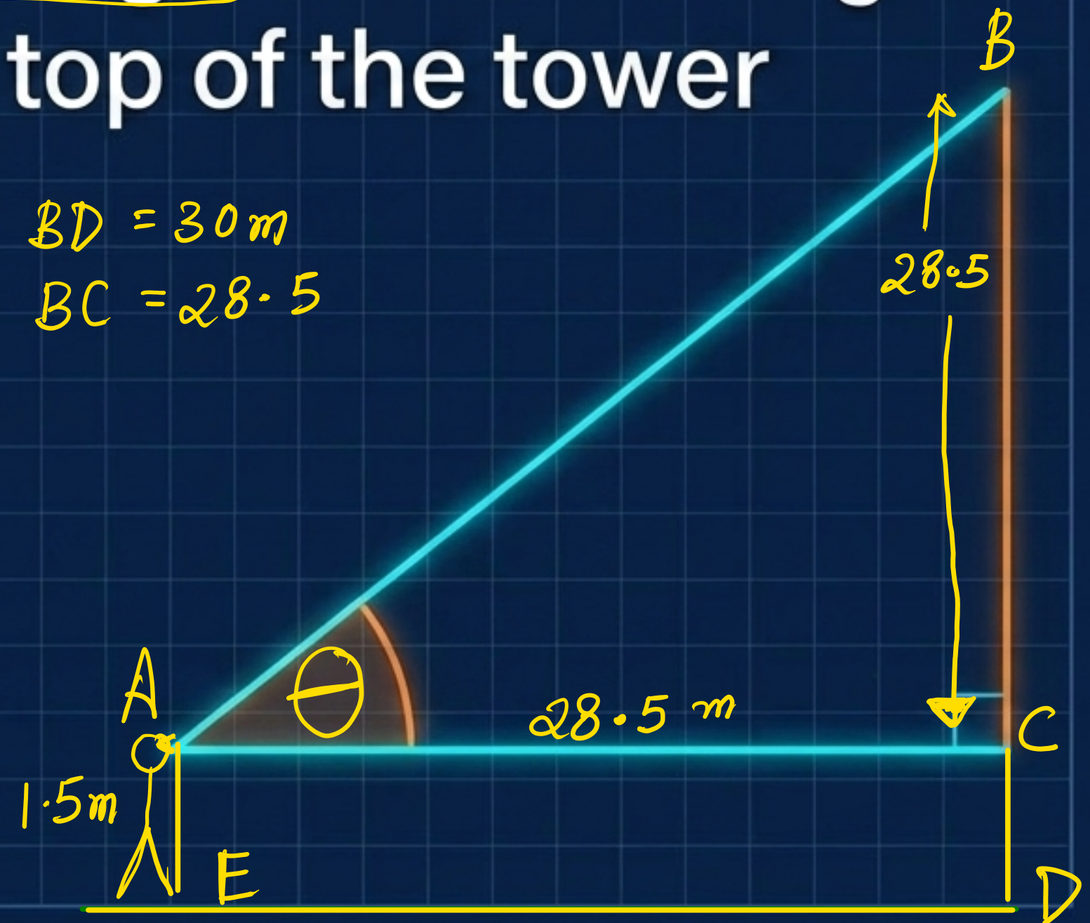
An observer, 1.5 m tall, is 28.5 m away from a tower 30 m high. Find the angle of elevation of the top of the tower from his eye.

$$\tan\theta = \frac{28.5}{28.5}$$

$$\tan\theta = 1$$

$$\theta = 45^\circ$$

BD = 30 m
BC = 28.5



[Practice Questions, Q2]

Correct Answer: 45°

Key Step: Effective height = 30 - 1.5 = 28.5m.
tan(θ) = 28.5 / 28.5 = 1.

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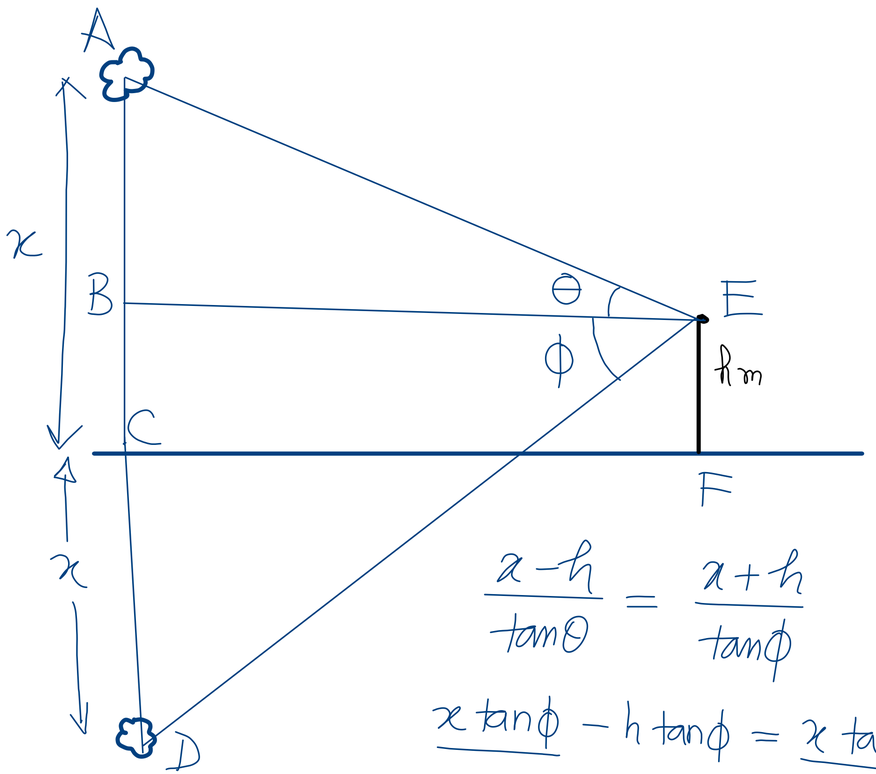
Sample Question 3 : The angle of elevation of a cloud from a point h metres above the surface of a lake is θ and the angle of depression of its reflection in the lake is ϕ.

Prove that the height of the cloud above the lake is $h\left(\dfrac{\tan\phi+\tan\theta}{\tan\phi-\tan\theta}\right)$.



In $\triangle ABE$

$\tan\theta = \dfrac{AB}{BE}$

$\tan\theta = \dfrac{x-h}{BE}$

$BE = \dfrac{x-h}{\tan\theta}$

In $\triangle DBE$

$\tan\phi = \dfrac{BD}{BE}$

$\tan\phi = \dfrac{x+h}{BE}$

$BE = \dfrac{x+h}{\tan\phi}$

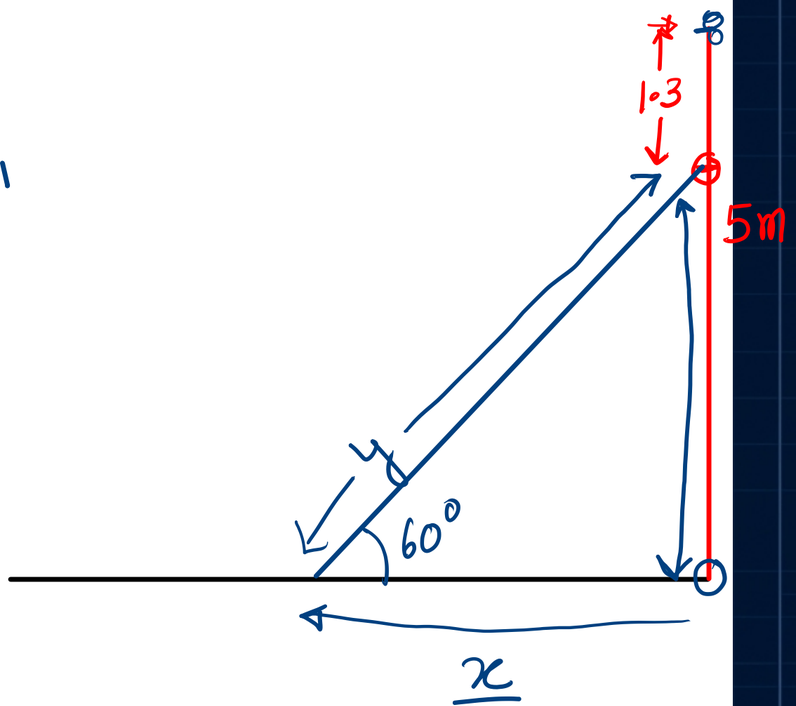
$$\frac{x-h}{\tan\theta} = \frac{x+h}{\tan\phi}$$

$$x\tan\phi - h\tan\phi = x\tan\theta + h\tan\theta$$

$$x\tan\phi - x\tan\theta = h\tan\theta + h\tan\phi$$

$$x(\tan\phi - \tan\theta) = h(\tan\theta + \tan\phi)$$

$$x = \frac{h(\tan\theta + \tan\phi)}{\tan\phi - \tan\theta}$$



Ascent Protocol 03: The Electrician's Repair

An electrician has to repair an electric fault on a pole of height 5 m. She needs to reach a point 1.3m below the top of the pole to undertake the repair work.

- **Task A:** What should be the length of the ladder inclined at 60°?
- **Task B:** How far from the foot of the pole should she place the foot of the ladder? (Take $\sqrt{3} = 1.73$)

[Practice Questions, Q15]

Correct Answer: Ladder ≈ 4.28 m | Distance ≈ 2.14 m
Key Step: Target height = 5 - 1.3 = 3.7m. $\sin(60°) = \sin(60°) = 3.7/\text{Ladder}$. $\cot(60°) = \text{Dist}/3.7$.

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Ascent Protocol 04: The Multi-Storeyed View

The angles of depression of the top and the bottom of an 8 m tall building from the top of a multi-storeyed building are 30° and 45°, respectively. Find the height of the multi-storeyed building and the distance between the two buildings.

[Practice Questions, Q19]

Correct Answer: Height = $4(3+\sqrt{3})$ m | Distance = $4(3+\sqrt{3})$ m

Key Step: $h/x = \tan(30°)$, $(h+8)/x = \tan(45°)$. $x = h+8$. $\sqrt{3}h = h+8$.

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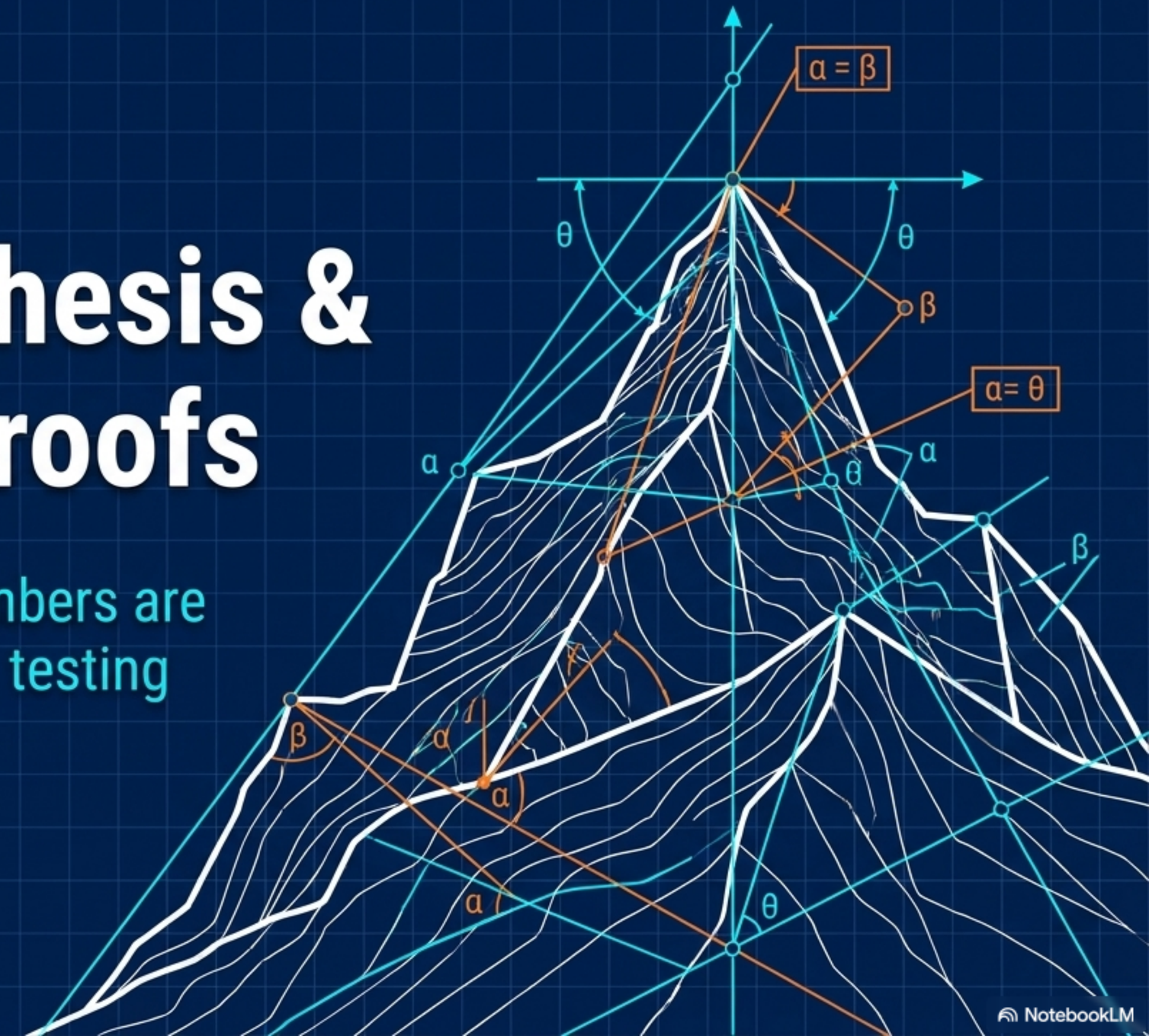
Phase III: The Summit

Variable Synthesis & Conceptual Proofs

Advanced problems where numbers are replaced by abstract variables, testing true structural understanding.



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Summit Challenge 01: Abstract Differences

The height of the tower is 100m. When the angle of elevation of the sun changes from 30° to 45°, the shadow of the tower becomes ‘x’ meters less. The value of ‘x’ is:

(a) $100\sqrt{3}$ m (b) 100 m

(c) $100(\sqrt{3}-1)$ m (d) $100/\sqrt{3}$ m

[MCQ III, Q4]

Correct Answer: (c) $100(\sqrt{3}-1)$ m

Key Step: Shadow at $45^\circ = 100$.

Shadow at $30^\circ = 100\sqrt{3}$.

Difference x = $100\sqrt{3}$ - 100.

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The angle of elevation of an aircraft from a point A on the ground is 60°. After a flight of 30 seconds, the angle of elevation changes to 30°. The aircraft is flying at a constant height of $3500\sqrt{3}$ m at a uniform speed. Find the speed of the aircraft. **5**

A drone was used to facilitate movement of an ambulance on the straight highway to a point P on the ground where there was an accident.

The ambulance was travelling at the speed of 60 km/h. The drone stopped at a point Q, 100 m vertically above the point P. The angle of depression of the ambulance was found to be 30° at a particular instant.

Based on above information, answer the following questions :

(i) Represent the above situation with the help of a diagram. **1**

(ii) Find the distance between the ambulance and the site of accident (P) at the particular instant. (Use $\sqrt{3} = 1.73$) **1**

(iii) (a) Find the time (in seconds) in which the angle of depression changes from 30° to 45°. **2**

OR

(iii) (b) How long (in seconds) will the ambulance take to reach point P from a point T on the highway such that angle of depression of the ambulance at T is 60° from the drone ? **2**

Summit Challenge 02: The Advancing Observer

The angle of elevation of the top of a tower from a point on the same level as the foot of the tower is α. On advancing 'p' meters towards the foot of the tower the angle of elevation becomes β.

- **Task A:** Show that the height 'h' of the tower is given by:

$$h = \frac{p \tan\alpha \tan\beta}{\tan\beta - \tan\alpha}$$

- **Task B:** Determine the height of the tower if $p = 150$ m, $\alpha = 30°$, and $\beta = 60°$.

[Practice Questions, Q46]

Correct Answer: Task B = $75\sqrt{3}$ m

Key Step: Base distance $x = h/\tan\beta$. Total distance $(x + p) = h/\tan\alpha$. $= h/\tan\alpha$. Substitute x to prove h.

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Summit Challenge 03: Reflection in the Lake

If the angle of elevation of a cloud from a point h metres above a lake is α and the angle of depression of its reflection in the lake be β...

Task: Prove that the distance of the cloud from the point of observation is:

$$\frac{2h \sec \alpha}{\tan \beta - \tan \alpha}$$

[Practice Questions, Q52]

Correct Answer: Proof verified by substitution.

Key Step: Cloud height H.
tan α = (H-h)/x, tan β = (H+h)/x.
Solve for x, equate. Dist = (H−h)/sin α.

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